

CS728: Assignment 1

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Chapter 1

Task 1

The following table shows the final parameters used for the GloVe training:

Hyperparameter	Value
Window Size (w)	10
Learning Rate (η)	0.05
Epochs	25
$x_{\max}$	100
α	0.75

Table 1.1: Selected GloVe hyperparameters

Nearest Neighbours for this task have been mentioned with results of Task 2.

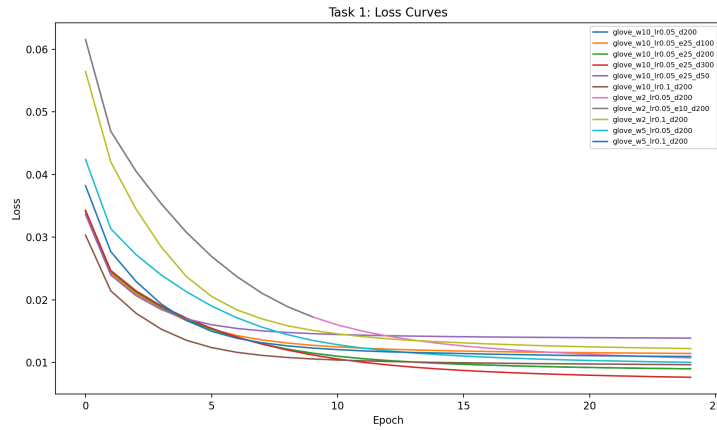


Figure 1.1: Loss curves for different embedding dimensions

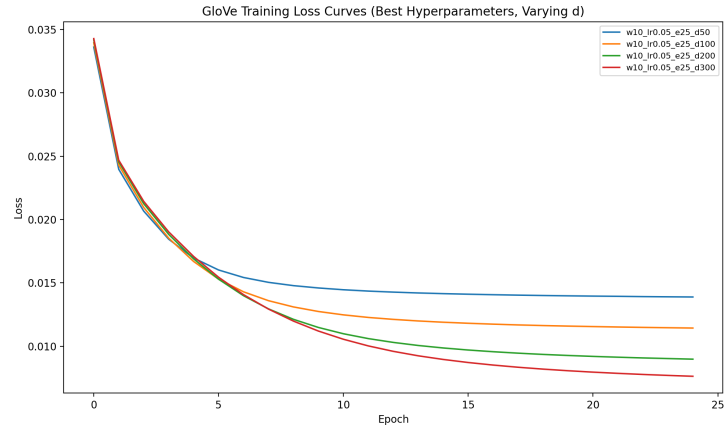


Figure 1.2: Loss curves for best hyperparameters across embedding dimensions

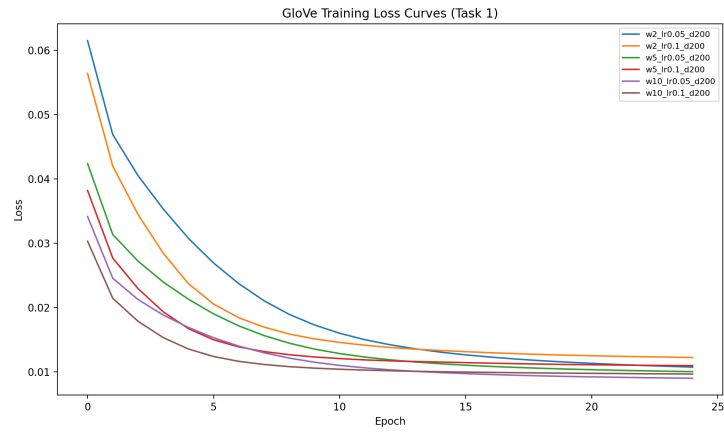


Figure 1.3: Loss curves for embedding dimension 200

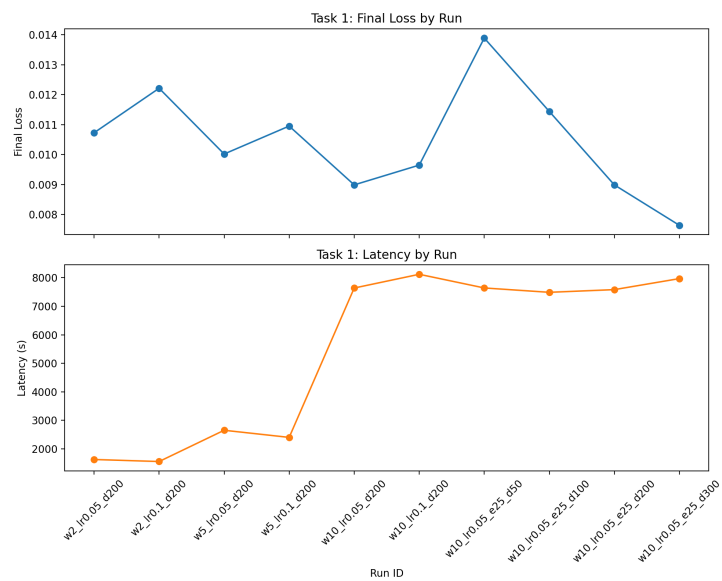


Figure 1.4: Final loss and Latency for different hyperparameters

Chapter 2

Task 2

Model	Word	Top-5 Neighbours
GloVe (d=200)	data	showed, statistics, tracking, according, information
	cities	towns, counties, across, areas, twin
	temperature	temperatures, maximum, levels, rainfall, room
SVD (d=50)	data	consumers, risk, tracking, technology, exposure
	cities	urban, influx, airports, communities, places
	temperature	heating, severe, fog, checking, causes
SVD (d=100)	data	automated, rapid, tool, specific, tracking
	cities	urban, transit, places, tourism, suburbs
	temperature	skin, heating, pump, cold, dry
SVD (d=200)	data	infrastructure, authentication, formulated, users, est
	cities	urban, mayor, city, transit, vast
	temperature	cold, slow, evaporate, brush, water
SVD (d=300)	data	mainframe, users, collected, infrastructure, breach
	cities	urban, city, mayor, vast, places
	temperature	cold, temperatures, evaporate, surface, fluids

Table 2.1: Top-5 nearest neighbours for selected words in GloVe and SVD models

Chapter 3

Task 3

The final parameters used for the CRF model were:

Parameter	Value
include_lexical	True
include_shape	True
include_subword	True
window_size	2

Table 3.1: CRF model parameters

Category	Features
Bias	bias
Lexical (Current Token)	word.lower, pos, chunk, pos[:2]
Lexical (Context ± 1)	-1:word.lower, -1:pos, -1:chunk +1:word.lower, +1:pos, +1:chunk
Lexical (Context ± 2)	-2:word.lower, -2:pos, -2:chunk +2:word.lower, +2:pos, +2:chunk
Shape (Current Token)	word.isupper, word.istitle, word.isdigit, word.shape, word.has_digit, word.has_hyphen, word.has_period
Shape (Context ± 1)	-1:word.isupper, -1:word.istitle, -1:word.isdigit, -1:word.shape +1:word.isupper, +1:word.istitle, +1:word.isdigit, +1:word.shape
Shape (Context ± 2)	-2:word.isupper, -2:word.istitle, -2:word.isdigit, -2:word.shape +2:word.isupper, +2:word.istitle, +2:word.isdigit, +2:word.shape
Subword	pref2, pref3, pref4, suf2, suf3, suf4
Sentence Boundary	BOS1, EOS1, BOS2, EOS2

Table 3.2: Complete feature set used with window size = 2

The table below lists the most important features learned:

Type	Feature	Weight
State	word.shape:x $\rightarrow$ O	5.9832
State	word.has_digit $\rightarrow$ O	5.2317
State	word.shape:x_x $\rightarrow$ O	4.8745
State	+2:word.lower=yellow $\rightarrow$ B-LOC	4.1123
State	-1:word.lower=v $\rightarrow$ B-ORG	3.9864
State	word.shape:Xx $\rightarrow$ B-PER	3.7421
State	pref3=mc $\rightarrow$ B-PER	3.6158
State	suf3=son $\rightarrow$ I-PER	3.5084
State	word.isupper=True $\rightarrow$ B-ORG	3.4219
State	word.istitle=True $\rightarrow$ B-PER	3.3376
Transition	B-PER $\rightarrow$ I-PER	6.4821
Transition	B-LOC $\rightarrow$ I-LOC	6.1734
Transition	B-ORG $\rightarrow$ I-ORG	5.9648
Transition	I-LOC $\rightarrow$ I-LOC	5.4127
Transition	O $\rightarrow$ I-PER	-6.2354
Transition	O $\rightarrow$ I-ORG	-6.0112
Transition	O $\rightarrow$ I-LOC	-5.8846
Transition	I-PER $\rightarrow$ I-PER	4.9731
Transition	I-ORG $\rightarrow$ I-ORG	4.8265
Transition	B-MISC $\rightarrow$ I-MISC	4.7029

Table 3.3: Top CRF features ranked by absolute weight magnitude.

Chapter 4

Task 4

We used a simple MLP architecture with a single hidden layer with 128 neurons followed by a ReLU and a dropout layer with a dropout rate of 0.2. The final output layer has 9 neurons corresponding to the 9 classes in the dataset. For OOV words we use the mean of the embeddings of the words in the training set as the embedding for the OOV word as it acts like a neutral embedding and does not bias the model towards any particular class. We trained the model for 10 epochs with a batch size of 1024 and a learning rate of 0.001 using the Adam optimizer. The tables below show the results of our model on the test set.

Dimension	Accuracy	Weighted F1	Macro F1
50	0.8309	0.7619	0.2179
100	0.8314	0.7643	0.2198
200	0.8314	0.7653	0.2224
300	0.8319	0.7658	0.2241

Table 4.1: Test performance of GloVe-MLP across embedding dimensions

Dimension	Accuracy	Weighted F1	Macro F1
50	0.8274	0.7523	0.2043
100	0.8301	0.7589	0.2091
200	0.8309	0.7609	0.2126
300	0.8305	0.7607	0.2118

Table 4.2: Test performance of SVD-MLP across embedding dimensions

From this we see that the best performance is achieved with the GloVe embeddings with a dimension of 300 and the SVD embeddings with a dimension of 200. This is worse than the CRF model which achieved macro F1 of 0.7983 and weighted F1 of 0.8342. This is because CRF model uses a window and hence has contextual information whereas and MLP trains on just the current word and does not have any context.

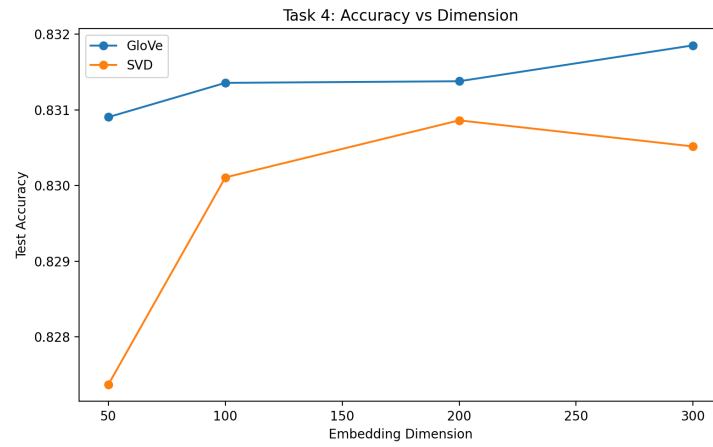


Figure 4.1: Test accuracy across embedding dimensions

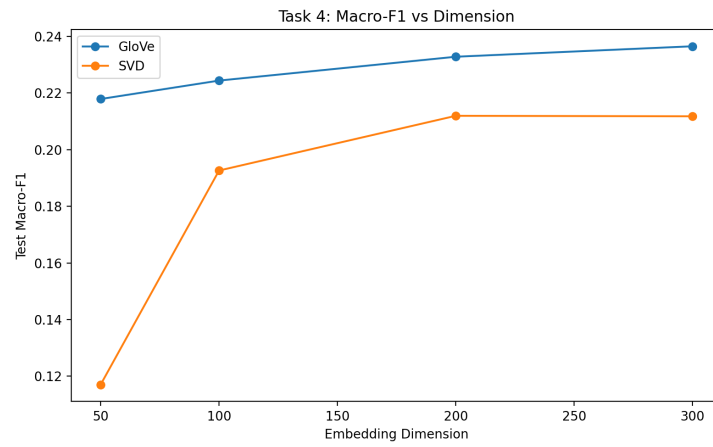


Figure 4.2: Test F1 score across embedding dimensions

Chapter 5

Task 5

Model	Word	Top-5 Neighbours
Raw SVD (d=200)	data	infrastructure, authentication, formulated, users, est
	temperature	cold, slow, evaporate, brush, water
	cities	urban, mayor, city, transit, vast
	apple	Not in vocabulary
	fast	enough, hard, keep, make, stopping
TFIDF SVD (d=200)	data	formulated, infrastructures, reads, est, mentioning
	temperature	cold, evaporate, temperatures, climatologist, gases
	cities	urban, inexorable, mayor, city, vast
	apple	Not in vocabulary
	fast	enough, hard, quick, make, up

Table 5.1: Comparison of Top-5 nearest neighbours for selected words in raw-SVD and TFIDF-SVD models

The MLP architecture was the same as the one used in task 4. The tables below show the results of our model on the test set.

Embedding	Accuracy	Weighted F1	Macro F1
SVD (Raw, d=200)	0.8287	0.7627	0.2163
SVD (TF-IDF, d=200)	0.8297	0.7625	0.2179

Table 5.2: Test performance comparison between Raw SVD and TF-IDF SVD embeddings (d=200)

We see that the TF-IDF embeddings perform marginally better than SVD.